

SOFTWARE ARCHITECT ASSESSMENT

Context: We are transitioning a critical business module from a legacy Java environment to a modern AI-orchestrated Python ecosystem. As the Architect, you are responsible for designing the bridge between our business logic and autonomous AI agents.

I. The Challenge

Problem: Build a tool that enables an AI Agent to calculate invoice taxes via the Model Context Protocol (MCP).

- The tool must calculate tax for a variable list of line items.
- The solution must be configurable to support future variations in regional logic without code changes.
- The system must provide an automated signal if the output requires human oversight (defined as tax exceeding \$40).
- The implementation must remain highly performant under load while maintaining a detailed audit trail.

II. Architectural Expectations

We are assessing your ability to build "Production-Ready" software. Your submission will be evaluated on:

- **Structural Integrity:** A decoupled design that allows for the transport layer (MCP) to be replaced with minimal friction.
- **Contract Safety:** Since this replaces a type-safe Java service, your implementation must enforce strict data integrity and validation.
- **Operational Readiness:** The system must be observable, traceable, and ready for containerized orchestration.
- **Agentic Readiness:** The tool definitions must be semantically rich enough for an LLM to reason about the logic and its outputs accurately.

III. Execution & Deployment

The service must be deployed to a public endpoint. During the review, an AI Agent will be connected to your live URL to perform a complex reasoning task based on your tool's output.

IV. Submission Requirements

- **Repository Link:** Project source code with a professional, scalable directory structure.
- **AI Context:** A comprehensive list of all prompts and instructions provided to AI tools during the development process, including any established governance rules.
- **Live URL:** The public endpoint for the MCP Server.
- **Design Document:** A README.md justifying your choice of patterns, your approach to observability, and how you resolved the conflict between auditing requirements and system performance.

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